# TOUBAL Mohamed Sahnoune

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## About Me

Third-year AI engineering student specializing in machine learning, computer vision, and NLP. Passionate about building end-to-end AI systems, from training complex **PyTorch** models to deploying highly scalable full-stack architectures and edge applications.

## Education

**ENSIA — National Higher School of Artificial Intelligence** *Expected 2028*
Third-Year Student
**Relevant Coursework**: Machine Learning, Data Mining, Mobile Dev, Web Dev, Data Structures & Algorithms I/II, AI, OOP, DB Systems, Linux.

## Technical Skills

**Core AI & Machine Learning Expertise**

- **Domains**: **Computer Vision** (Image Classification, Face Recognition, CNNs), **Natural Language Processing** (**RAG** Pipelines, LLMs), **Federated Learning**.
- **Data & ML Engineering**: End-to-End Pipeline Architecture, Data Preprocessing, Feature Engineering.
- **Frameworks & Libraries**: **PyTorch**, Scikit-learn, Pandas, NumPy.

**Software Engineering & Infrastructure**

- **Architecture & Edge**: Clean Architecture (BLoC/Cubit, Repository Pattern), **On-Device AI Inference**.
- **Languages**: Python, C++, Dart, JavaScript/TypeScript, SQL, Bash.
- **Full-Stack & Tools**: **Flutter**, React, Next.js, Express, PostgreSQL, **PostGIS**, Git, Linux.

## Projects

### Ransomware Detection and Mitigation System *2026*

- Developed a **research-grade ransomware detection pipeline** using **70 engineered memory-forensics features**, hybrid ML models, and an ensemble targeting **93–95% recall** and **98% ROC-AUC**.
- Built an **automated mitigation engine** that standardizes model alerts and triggers responses such as **process suspension, termination, network blocking, notifications, and event logging**.

### StreetWatch: AI-Powered Road Damage Reporter *2026*

- Developed a patch-based **CNN** for image classification, achieving **94% accuracy** with a 2-second inference time on mobile devices.
- Implemented clean architecture (Repository pattern, BLoC) in **Flutter** for robust **on-device AI** integration.
- Built a web dashboard with **PostGIS** for spatial analysis and real-time mapping of reported infrastructure issues.

### Fantasy Premier League Prediction Pipeline *2025 – 2026*

- Engineered an end-to-end pipeline using 10 seasons of match data to predict player performance with a Mean Absolute Error (**MAE**) of **1.2**.
- Performed advanced **feature engineering** and data merging to integrate team context and match-day metrics.

### Drone Based Delivery Optimization Problem *2026*

- Developed a drone-based delivery optimization project focused on generating reproducible routing datasets and instance-based logistics scenarios to support route planning, fleet allocation, and optimization analysis for last-mile delivery systems

## Languages

Arabic (Native), English (Professional), French (Professional (TCF Score : C2))